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Berdakh Abibullaev, Ph.D.

Curriculum Vitae

Research Area

Neural Engineering: Brain-Computer/Machine Interfaces; EEG/MEG/ECoG/fNIRS signal processing.

Machine Learning: Neural Networks and Deep Learning, Kernel based learning, Support vector machines.

Education

02/2010 **Ph.D**, *Yeungnam University*, Department of Electronic Engineering, South Korea.
- *Research area* : EEG time-series prediction, Machine learning and Neural Networks;

08/2006 **M.Sc**, *Yeungnam University*, Department of Electronic Engineering, South Korea.
- *Research area*: Wavelet Analysis, Time-frequency localization; Physiological feature extraction;

07/2004 **B.Sc**, *Tashkent University of Information Technologies*, Uzbekistan.

Academic Work Experience

08/2022-present **Associate Professor**, *Robotics & Mechatronics Department, School of Engineering and Digital Science, Nazarbayev University*, Astana, Kazakhstan.

Research Focus:
- Developing advanced machine learning and signal processing strategies, with a focal point on creating interfaces that facilitate real-time communication between the brain and computational devices.
- Optimizing deep learning techniques with the goal of creating universally applicable models that go beyond individual and session-specific differences in Brain-Computer/Machine Interfaces.

Currently Teaching:
- Conducting a detailed course on the practical applications of "Machine Learning," blending theoretical insights with hands-on experience
- Leading instructive sessions on "Microcontrollers," providing both conceptual knowledge and practical skills.
- Guiding students through an all-encompassing curriculum on "Brain-Machine Interfaces," highlighting the convergence of neuroscience and technology.

09/2015-08/2022 **Assistant Professor**, *Robotics & Mechatronics Department, School of Engineering and Digital Science, Nazarbayev University*, Astana, Kazakhstan.

Teaching Experience:
- Instructed "Signals and Sensing," incorporating hands-on laboratory sessions.
- Led the course "Machine Learning: Practical Applications."
- Delivered both theoretical and practical knowledge in "Microcontrollers."
- Developed and taught a comprehensive curriculum on "Brain-Machine Interfaces."

Research Focus:
- Expertise in devising machine learning and signal processing solutions, specifically for real-time Brain-Computer/Machine Interfaces.
- Advanced the use of deep learning models to achieve universal applicability across different sessions and subjects in Brain-Computer/Machine Interfaces.

5/2018 - 7/2018 **Visiting Professor**, *Department of Electrical Engineering & Computer Science, University of Houston*, Houston, Texas, USA.
- Cortical source imaging of electroencephalographic signals in patients with stroke to examine neural correlates of brain plasticity pre and post-clinical trials with BMI actuated exoskeleton.

9/2014 - 9/2015 **NIH Postdoctoral Research Fellow II**, *Department of Electrical Engineering & Computer Science, University of Houston*, Houston, Texas, USA.
- BMI-Exo Project: Control of Therapeutic Exoskeletons through Brain-Machine Interface
  - Conducted research on developing interfaces that capture brain signals indicating intent, facilitating the movement of a therapeutic exoskeleton for patient rehabilitation.
  - Explored the underlying brain functions responsible for effective BMI interactions.
- Investigation of Infant Mirror Neuron System
  - Led research on the mirror neuron system's influence on infants' motor learning using non-invasive brain data collection methods.
  - Utilized broad brain connectivity assessments to gain a clearer understanding of neural pathways influencing infant behavioral patterns.
- Electromagnetic Source Localization in the Brain
  - Developed methods for accurately locating EEG sources from brain activity.
  - Applied and refined connectivity analysis techniques to map and understand complex brain network patterns.

01/2014-09/2014 **Research Scientist**, *Department of Neurology, Samsung Medical Center*, Seoul, South Korea.

- Analyzed High-Frequency Oscillations in ECoG, EEG, enhancing the understanding of Epileptogenesis.
- Developed advanced machine learning techniques for in-depth ECoG, EEG data analysis.
- Worked with EEG & MEG recordings to pinpoint epileptogenic zones, aiding in refining surgical treatments for refractory focal epilepsy.
- Collaborated with medical teams, aiding in presurgical evaluations for patients with treatment-resistant epilepsy.
- Leveraged my ECoG/EEG expertise to provide insights that improved patient treatment plans.

04/2011-12/2013 **Senior Research Scientist**, *Robotics Research Division, Daegu Gyeongbuk Institute of Science and Technology*, South Korea.

- Worked with kernel-based learning algorithms for BCIs, aiding in the control of neuro-rehabilitation devices.
- Explored inter-session and inter-subject variabilities, contributing to better fNIRS-based BCI performance.
- Assisted in enhancing non-invasive BCI paradigms, including ERP-based and Sensory-Motor Rhythms-based BCIs.
- Contributed to a team focused on creating a Hybrid BCI system, which integrates brain hemodynamic and electrical signals.

01/2010-04/2011 **Postdoctoral Research Scholar**, *Robotics Research Division, Daegu Gyeongbuk Institute of Science and Technology*, South Korea.

- Collaborated on using Brain-Computer Interfaces based Neurofeedback system to support therapy and diagnosis of Attention Deficit and Hyperactivity Disorders (ADHD).
- Contributed to the application of quantitative EEG and Neurofeedback methods tailored for children with ADHD.
- Assisted in identifying key neural signal patterns in children with ADHD.
- Worked on the development of semi-supervised feature selection and classification algorithms.
- Engaged in the long-term developmental assessment of children with ADHD in partnership with the neurorehabilitation center at Daegu University.

09/2006 - 02/2010 **Research Assistant**, *Research Assistant, Department of Electronic Engineering, Yeungnam University*, South Korea.

- Worked on spectral analysis methods to pinpoint neural signal patterns linked to Epileptic disorders.
- Investigated variabilities in neural activity across patients, focusing on distinguishing seizure from non-seizure patterns to predict seizure onset.
- Engaged in a collaborative partnership with physicians at Yeungnam University and Keimyung University Medical Center. My role encompassed long-term EEG data collection, video monitoring of patients experiencing epileptic seizures, and in-depth EEG data analysis.
- Delved into classifying sleep stage disorders using EEGs, aiming for early identification of sleep apnea, a nocturnal respiratory condition.

Honors and Awards

06/2014 The Best Paper Award for fNIRS motion artifact correction for overground walking using entropy-based unbalanced optode decision and wavelet regression neural network in Fusion and Integration for Intelligent Systems (MFI), 2017 IEEE International Conference, South Korea.

12/2012 The Best Poster Award for The evaluation of accuracy and clinical usefulness of 3D EEG source localization analysis at the Korean Epilepsy Congress 2014.

12/2012 The Best Researcher of the Year Award for the Robotics Research Division at Daegu Gyeongbuk Institute of Science and Technology, South Korea.

09/2006 Korean Government Scholarship recipient of the competitive Korean Government Information Technology Scholarship for International Graduate Students for a Ph.D. program, South Korea.

09/2004 Brain Korea Scholarship recipient of the competitive graduate scholarship for an M.Sc. program in South Korea.

Technical Skills

Software skills.

Expert in Python and libraries such as Cython. Good knowledge of C and C++. Experience with various tools and languages, including bash, LaTeX, HTML, Git, and database query languages

Data Science.

Expert in NumPy, Scipy, Scikit-learn, PyTorch, TensorFlow, and Azure DevOps.

Neural data acquisition systems.

Experienced user of Nicolet EEG Amplifiers, Blackrock Microsystems for intracranial recording, Functional Near-Infrared Spectroscopy FOIRE3000, BIOPAC fNIR200 Optical Brain Imaging, BioSemi Active-Two EEG system, Brain Products EEG amp., BIOPAC MP100 EEG amp., and G.tec g.USBamp systems.

Productivity applications.

TEX: [LATEX, BibTEX, PSTricks, Beamer].

System Environments.

Linux (Pop_OS, Ubuntu), Windows

Service to the Scientific Community

Associate Editor.

- IEEE Access Journal (Impact Factor: 3.7)
- PeerJ Computer Science (Impact Factor: 3.09)

Occasional Reviewer.

- IEEE Access Journal
- Neuroimage, Elsevier.
- IEEE Transactions on Neural Networks and Learning Systems
- Brain Research, Elsevier.

Program Committee Member and Session Chair.

- International Workshop on Artificial Intelligence for Healthcare Applications at the International Conference on Pattern Recognition 2020, Italy.
- The 7th International Conference on New Trends in Information Science, Service Science, and Data Mining, June 18-20, 2013, Jeju Island, South Korea.
- Session Chair: IEEE International Conference on Multisensory Fusion and Integration for Intelligent Systems, 2017, Daegu, South Korea.
- The 8th International Conference on Information Processing, Management, and Intelligent Information Technology, April 1-3, 2013, Seoul, South Korea.
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Project Proposals, Participation & Coordination

01/2021 **Principal Investigator**, ***Brain-Machine Interfaces for Neural Rehabilitation after Stroke***, This project aims to accelerate the efficacy of robotic rehabilitation using a new neural interface system developed for upper-limb repair after stroke. It aims to facilitate expediate motor recovery from incomplete upper-limb paralysis and investigate in-depth understanding of cortical reorganization in the brain associated with the recovery..

01/2021 **Co-Investigator**, ***Classifier Validity and Its Applications in Bioinformatics***, The project studies the measures of classifier validity and explores their applications in bioinformatics. Nazarbayev University..

01/2019 **Co-Investigator**, ***Neural-assisted optimization of human learning with multimodal neuroimaging of EEG and NIRS***, Implementing and validate a neural-assisted, machine-learning-based intelligent tutoring system that uses information from physiological signals to adapt the learning schedule to a user's individual learning capacity..

09/2015 **Principal Investigator**, ***Development of a Brain-Computer Interface with a Robot and Functional-Electrical Stimulation for Upper-Limb Stroke Rehabilitation (BCI-ROBOFES)***, Evaluated by the committee of Oak Ridge Associated Universities (ORAU);.

9/2014 **Participant**, *Cognitive-Motor Development Research in Infants.*, Lead a team of Ph.D. students in exploring & decoding of the neural basis of *Imitation and Action Understanding* in freely-behaving infants via multimodal brain imaging techniques, University of Houston, Texas.

9/2014 **Participant**, ***The BMI-Exo Project: Facilitating Personalized Robotic Rehabilitation Of The Upper Limb, [NIH Grant R01NS081854-02, National Robotics Initiative]***, Lead a part of research that focuses on understanding neurophysiological correlates of motor learning during movements in stroke patients..

04/2014 **Participant**, ***Development of high-resolution subdural electrodes and data mining methods for precise localization of epileptogenic zone in treatment of patients with refractory epilepsy.***, Participated in idea generation and in drafting/writing of the project proposal at Samsung Medical Center, Korea.

Publications

Patents

[1] J. An, S.H. Jin, S.H. Lee, J.I. Moon B. Abibullaev, J.H. Ahn and G.H. Jang. **Rehabilitation Training System and Method**. *U.S. Patent No*: 9,081,890,Washington, DC: United States. Patent and Trademark Office, 2015 (https://www.google.com/patents/US9081890).

[2] J. An, S.H. Jin, S.H. Lee, J.I. Moon B. Abibullaev, J.H. Ahn and G.H. Jang. **Self-directed rehabilitation training method combining brain signals and functional electro-stimulation**. *EFS ID*: 17077057, 09-OCT-2013, United States, (http://www.google.com/patents/US20140200632).

[3] J. An, S.H. Jin, S.H. Lee, B. Abibullaev, J.I. Moon , J.H. Ahn and G.H. Jang. **Combining Brain Signals and Functional Electrostimulation Self-Directed Rehabilitation Method**, Patent Registration Number: 101501524000, 05.03.2015. South Korea,(http://eng.kipris.or.kr/).

Refereed Journal Publications

[1] B. Abibullaev, Kassymzhomart Kunanbayev and A. Zollanvari, **Subject-Independent Classification of P300 Event-Related Potentials Using a Small Number of Training Subjects**, *IEEE Transactions on Human-Machine Systems*, DOI: 10.1109/THMS.2022.3189576, 2022.(https://doi.org/10.1109/THMS.2022.3189576)***(Impact Factor: 4.124**; Quartile rank: Q1).*

[2] I. Dolzhikova, B. Abibullaev, R. Sameni and A. Zollanvari, **Subject-Independent Classification of Motor Imagery Tasks in EEG using Multi-Subject Ensemble CNN**, *IEEE Access*, DOI: 10.1109/ACCESS.2022.3195513, 2022.(https://doi.org/10.1109/ACCESS.2022.3195513)***(Impact Factor: 3.476**; Quartile rank: Q1).*

[3] B. Abibullaev and A. Zollanvari, **A Systematic Deep Learning Model Selection for P300-Based Brain-Computer Interfaces**, *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, DOI: 10.1109/TSMC.2021.3051136, 2021.(https://ieeexplore.ieee.org/document/9351887)***(Impact Factor: 13.451**; Quartile rank: Q1).*

[4] A. Zollanvari and B. Abibullaev **Bias Correction for Linear Discriminant Analysis**, *Pattern Recognition Letters*, vol. 151, pp. 41-47, 2021, https://doi.org/10.1016/j.patrec.2021.07.026 *(Impact Factor: 3.7; Quartile rank: Q1).*

[5] B. Abibullaev and A. Zollanvari, **Learning Discriminative Spatiospectral Features of ERPs for Accurate Brain-Computer Interfaces** *IEEE Journal of Biomedical and Health Informatics*, vol. 98, pp. 1-12, 2019, DOI: 10.1109/JBHI.2018.2883458 *(Impact Factor: 5.772; Quartile rank: Q1).*

[6] B. Abibullaev, I. Dolzhikova and A. Zollanvari, **A Brute-force CNN Model Selection for Accurate Classification of Sensorimotor Rhythms in BCIs** *IEEE Access*, 2020, DOI: 10.1109/ACCESS.2020.2997681 *(Impact Factor: 3.36; Quartile rank: Q1).*

[7] B. Abibullaev, A. Zollanvari, B. Saduanov and T. Alizadeh, **Design and Optimization of a BCI-Driven Telepresence Robot Through Programming by Demonstration** *IEEE Access*, August 2019, DOI: 10.1109/ACCESS.2019.2933268 *(Impact Factor: 3.36; Quartile rank: Q1).*

[8] A. Zollanvari, M. Abdirash, A. Dadlani and B Abibullaev. **Asymptotically Bias-Corrected Regularized Linear Discriminant Analysis for Cost-Sensitive Binary Classification** *IEEE Signal Processing Letters*, 22 May 2019, DOI: 10.1109/LSP.2019.2918485 *(Impact Factor: 3.1; Quartile rank: Q1).*

[9] B. Abibullaev, J An, SH Lee, JI Moon. **Design and Evaluation of Action Observation and Motor Imagery based BCIs using NIRS** *Measurement*,vol. 98, pp. 250-261, 2017, Elsevier *(Impact Factor: 3.927; Quartile rank: Q1).*

[10] N.A. Bhagat, A. Venkatakrishnan, B. Abibullaev, E.J. Artz, N. Yozbatiran, A. Blank, J. French, C. Karmonik, R.G.Grossman, M.K O'Malley, G. Francisco, J.L. Contreras-Vidal. **Design and optimization of an EEG-based brain-machine interface (BMI) to an upper-limb exoskeleton for stroke survivors.** *Frontiers in Neuroscience*, vol. 10, March, 2016 (Impact Factor: 4.677; Quartile rank: Q1).

[11] J.G. Cruz-Garza, Z.R. Hernandez, T. Tse, E. Caducoy, B. Abibullaev, J.L. Contreras-Vidal. **A novel experimental and analytical approach to the multimodal neural decoding of intent during social interaction in freely-behaving human infants.** *Journal of Visualized Experiments*, DOI: 10.3791/53406, October, 2015 (Impact Factor : 1.5; Quartile rank: Q2).

[12] C.H. Park, J.H Seo, D. Kim, B. Abibullaev, H. Kwon, Y.H. Lee, M.Y. Kim, K. Kim, J.S. Kim, E.Y. Joo, S.B. Hong, (2015, Feb). **EEG Source Imaging in Partial Epilepsy in Comparison with Presurgical Evaluation and Magnetoencephalography.** *Journal of Clinical Neurology*, 2015 Feb 17, 11:e12 (Impact Factor : 3.077; Quartile rank: Q2).

[13] B. Abibullaev, J An, S.H. Jin, and J.I. Moon. **Classification of brain hemodynamic signals arising from visual action observation tasks for brain-computer interfaces : An fNIRS study.** *Measurement*, DOI:10.1016/j.measurement.2013.11.044, 2014. Elsevier (Impact Factor : 3.927; Quartile rank: Q1).

[14] B. Abibullaev, J An, S.H. Lee, S.H. Jin, and J.I. Moon. **Minimizing inter-subject variability in fNIRS based brain computer interfaces via multiple-kernel support vector learning.** DOI: 10.1016/j.medengphy.2013.08.009, *Medical Engineering Physics*, 2013. Elsevier (Impact Factor : 2.242; Quartile rank: Q2).

[15] B. Abibullaev and J. An. **Classification of frontal cortex hemodynamic response during cognitive tasks using wavelet transforms and machine learning algorithms.** *Medical Engineering Physics*, 34(10):1394–410, 2012. Elsevier (Impact Factor : 2.242; Quartile rank: Q2).

[16] B. Abibullaev and J. An. **Decision support algorithm for diagnosis of ADHD disorder using electroencephalograms.** *Journal of Medical Systems*, 36(4):2675–2688, 2011. Springer (Impact Factor : 4.460; Quartile rank: Q2).

[17] B. Abibullaev, J. An, and J.I. Moon. **Neural network classification of brain hemodynamic responses from four mental tasks.** *Int J Optomechatronics*, 5(4):340–359, 2011. Taylor & Francis (Impact Factor : 1.9; Quartile rank: Q2).

[18] B. Abibullaev and H.D. Seo. **A new QRS detection method using wavelets and artificial neural networks.** *Journal of Medical Systems*, 35(4):683–691, 2011. Springer (Impact Factor : 4.460; Quartile rank: Q2).

[19] B. Abibullaev, M.S. Kim, and H.D. Seo. **Epileptic spike detection using continuous wavelet transforms and artificial neural networks.** *Int J of Wavelets, Multires and Inform Proc*, 8(1):33–48, 2010. Worldscientific (Impact Factor : 0.463; Quartile rank: Q3).

[20] B. Abibullaev, M.S. Kim, and H.D. Seo. **Seizure detection in temporal lobe epileptic EEGs using the best basis wavelet functions.** *Journal of Medical Systems*, 34(4):755–765, 2010. Springer (Impact Factor : 4.460; Quartile rank: Q2).

[21] M.S. Kim, Y.C. Cho, B. Abibullaev, and H.D. Seo. **Analysis of brain functions and classification of sleep stage EEG using daubechies wavelet.** *Sensors and Materials*, 20(1):1–15, 2008. MYU, Japan (Impact Factor: 0.92; Quartile rank: Q3).

Conference proceedings

[1] V. Mun And B. Abibullaev, "**Explainable Deep Learning for Brain-Computer Interfaces through Layerwise Relevance Propagation**", 2023 11th International Winter Conference on Brain-Computer Interface (BCI), 2023, pp. 1-5, doi: 10.1109/BCI53720.2022.9734904.

[2] I. Dolzhikova, B. Abibullaev, A. Zollanvari, "**An Ensemble of Convolutional Neural Networks for Zero-Calibration ERP-Based BCIs**", 2022 10th International Winter Conference on Brain-Computer Interface (BCI), 2022, pp. 1-4, doi: 10.1109/BCI53720.2022.9734904.

[3] I. Dolzhikova, B. Abibullaev, R. Sameni, A. Zollanvari, "**An Ensemble CNN for Subject-Independent Classification of Motor Imagery-based EEG**", the 43rd Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBS), October 31 – November 4, 2021

[4] Y. Kainolda, B. Abibullaev, R. Sameni, A. Zollanvari, "**Is Riemannian Geometry Better than Euclidean in Averaging Covariance Matrices for CSP-based Subject-Independent Classification of Motor Imagery?**", the 43rd Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBS), October 31 – November 4, 2021

[5] M. Nurpeiissov, B. Abibullaev, T. Alizadeh, "**A novel human-robot interaction framework based on Telegram and programming by demonstration**", 9th International Conference on Robot Intelligence Technology and Applications (RITA 2021), on Dec 16 – 17, 2021 at KAIST Daejeon, Korea.

[6] S. Abilkassov, M. Kairgaliyev, B. Zhakanov, B. Abibullaev, "**A System For Drivers' Cognitive Load Estimation Based On Deep Convolutional Neural Networks and Facial Feature Analysis**", in 22nd IEEE International Conference on Industrial Technology, Virtual 10-12, 2021, Spain.

[7] K. Kunanbayev, B. Abibullaev, A. Zollanvari, "**Data Augmentation for P300-based Brain-Computer Interfaces Using Generative Adversarial Networks**", in 2021 9th International Winter Conference on Brain-Computer Interface (BCI), 22-24 Feb. 2021, pp. 1-7, Gangwon, Korea (South), doi: 10.1109/BCI51272.2021.9385317.

[8] K. Kunanbayev, D. Azhigulov, B. Abibullaev, A. Zollanvari, "**Deep Transfer Learning for Subject-Independent ERP-based BCIs**", in 2021 9th International Winter Conference on Brain-Computer Interface (BCI), 22-24 Feb. 2021, pp. 1-3, Gangwon, Korea (South), doi: 10.1109/BCI51272.2021.9385311.

[9] A. Oleinikov, B. Abibullaev, M. Folgheraiter, "**On the Classification of Electromyography Signals to Control a Four Degree-Of-Freedom Prosthetic Device**," in 2020 42nd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC), July 20-24, Montreal, Canada.

[10] B. Saduanov, D. Tokmurzina, K. Kunanbayev and B. Abibullaev, "**Design and Optimization of a Real-Time Asynchronous BCI Control Strategy for Robotic Manipulator Assistance**", in 2020 8th International Winter Conference on Brain-Computer Interface. (BCI), 2020, GangWon, South Korea.

[11] Adilet Tuleuov and B. Abibullaev, "**Deep Learning Models for Subject-Independent ERP-based Brain-Computer Interfaces**," in *the 2019 9th International IEEE/EMBS Conference on Neural Engineering (NER)*, March 20-23, 2019, San Francisco, CA, USA.

[12] B. Abibullaev, Y. Orazayev, and A. Zollanvari, "**Novel Spatiospectral Features of ERPs Enhances Brain-Computer Interfaces**," in *Brain-Computer Interface (BCI), 2019 7th International Conference on*. IEEE Brain Society, 2019, GangWon, South Korea.

[13] B. Saduanov, D. Tokmurzina, T. Alizadeh, and B. Abibullaev, "**Brain-computer interface humanoid pre-trained for interaction with people**," in *2018 ACM/IEEE International Conference on Human-Robot Interaction*. ACM, March 5-8, 2018, pp. 229–230, Chicago, IL, USA.

[14] A. Oleinikov, B. Abibullaev, A. Shintemirov, and M. Folgheraiter, "**Feature extraction and real-time recognition of hand motion intentions from emgs via artificial neural networks**," in *Brain-Computer Interface (BCI), 2018 6th International Conference on*. IEEE Brain Society, 2018, pp. 1-5, GangWon, South Korea.

[15] G. Lee, S. H. Jin, S. T. Yang, J. An, and B. Abibullaev, "**Cross-correlation between HBO and HbR as an effective feature of motion artifact in the fNIRS signal**," 2018 6th International Conference on Brain-Computer Interface (BCI), pp. 1-3, 2018, IEEE Brain Society, GangWon, South Korea.

[16] B. Saduanov, T. Alizadeh, J. An, and B. Abibullaev, "**Trained by demonstration humanoid robot controlled via a BCI system for telepresence**," in *Brain-Computer Interface (BCI), 2018 6th International Conference on*. IEEE Brain Society, 2018, pp. 1–4, January, GangWon, South Korea.

[17] G. Lee, S. H. Jin, S. H. Lee, B. Abibullaev, and J. An, "**fNIRS motion artifact correction for overground walking using entropy-based unbalanced optode decision and wavelet regression neural network**," in *Multisensor Fusion and Integration for Intelligent Systems (M.F.I.), 2017 IEEE International Conference on*. IEEE, 2017, pp. 186–193, Daegu, South Korea.

[18] A. Zhumadilova, D. Tokmurzina, A. Kuderbekov, and B. Abibullaev. **Design and Evaluation of a P300 Visual Brain-Computer Interface Speller in Cyrillic Characters**. In *the 26th IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN), IEEE*, 2017, Aug 28 - Sept 1, Lisbon, Portugal.

[19] B. Abibullaev. **Learning Suite of Kernel Feature Spaces Enhances SMR-Based EEG-BCI Classification.** In *the 5th International Winter Conference on Brain-Computer Interface, IEEE*, 2017, January 9-11, GangWon, South Korea.

[20] D. Nurseitov, A. Serekov, A. Shintemirov and B. Abibullaev. **Design and Evaluation of a P300-ERP based BCI System for Real-Time Control of a Mobile Robot.** In *the 5th International Winter Conference on Brain-Computer Interface, IEEE*, 2017, January 9-11, Gangwon , South Korea.

[21] B. Abibullaev and J An. **On Robust Classification of Hemodynamic Signals for BCIs via Multiple Kernel ν-SVMs.** *Intelligent Robots and Systems (IROS), 2016 IEEE/RSJ International Conference*, 2016, Oct10-15, Daejeon, South Korea.

[22] J.G. Cruz-Garza, Z.R. Hernandez, M. Megjhani, B. Abibullaev, T. Tse, E. Caducoy, and JL Contreras-Vidal. **Neural Development of social cognition in the first two years of life - Early findings.** In *Society for Neuroscience* 2015 Oct 17-21; Chicago, USA.

[23] Z.R. Hernandez, A.J. Arenas-Castellanos, J.G. Cruz-Garza, M. Megjhani, B. Abibullaev, Sri. R.P. Maddi, T. Tse, C. Armstrong, W. Long, J.L. Contreras-Vidal **Decoding Intent From Non-invasive EEG in Freely Behaving Infants.** In *Ninth Biennial Meeting of the Cognitive Development Society* 2015 Oct 9-10; Ohio, U.S.A.

[24] A.J. Arenas-Castellanos, Z.R. Hernandez, J.G. Cruz-Garza, M. Megjhani, B. Abibullaev, Sri. R.P. Maddi, T. Tse, C. Armstrong, W. Long, J.L. Contreras-Vidal **A developmental Analysis of Behaviors Related to the Mirror Neuron System in 6-24 Months Infants** In *Ninth Biennial Meeting of the Cognitive Development Society* 2015 Oct 9-10; Ohio, U.S.A.

[25] Z.R. Hernandez, J.G. Cruz-Garza, T. Tse, E. Caducoy, B. Abibullaev, J.L. Contreras-Vidal. **Supervised Classification of Intended Behaviors Using Electroencephalography (EEG) from Freely-Behaving Infants: Early Findings.** In *12th Annual G.C.C. Conference on Theoretical and Computational Neuroscience*, Rice University, Houston, TX, United States, February 6-7, 2015.

[26] N. A. Bhagat, A Venkatakrishnan, B. Abibullaev, E. J. Artz, A.A. Blank, J. A. French, N. Yozbatiran, and R. G. Grossman, M. K. O'Malley, J. L. Contreras-Vidal, G. E. Francisco **BMI Control of a Therapeutic Exoskeleton to Facilitate Personalized Robotic Rehabilitation of the Upper Limb.** *In National Robotics Initiative (N.R.I.) P.I. Meeting*, Westin Arlington, Nov. 19-20 2014, United States.

[27] C.H.Park, B. Abibullaev, E.Y. Joo, S.C. Hong and S.B. Hong. **The evaluation of accuracy and clinical usefulness of 3D EEG source localization analysis.** In *the 19th Korean Epilepsy Congress*, Seoul, Republic of Korea, June 12-14 2014.

[28] S.H. Lee, J. An, G. Jang, S.H. Jin, B. Abibullaev, and J.I. Moon. **Neural activity during observation, imagery, and execution of eating: An fNIRS pilot study.** In *the 19th Annual Meeting of the Organization for Human Brain Mapping*, Seattle, WA, United States, June 16-20, 2013.

[29] J. An, S.H. Jin, S.H. Lee, G. Jang, B. Abibullaev, J. Ahn, H. Lee, and J.I. Moon. **Cortical activation pattern for grasping during observation, imagery, execution, FES, and observation-FES integrated BCI: An fNIRS pilot study.** In *the 35th Annual Int. Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, Osaka, Japan, July 3-7 2013.

[30] J. An, S.H. Lee, S.H. Jin, B. Abibullaev, G. Jang, J. Ahn, H. Lee, and J.I. Moon. **The beginning of neurohaptics: Controlling cognitive interaction via brain haptic interface.** In *the 2013 IEEE International Winter Workshop on Brain-Computer Interface*, Gangwon Province, Korea, Feb. 18-20 2013.

[31] B. Abibullaev, J. An, J.I. Moon, S.H. Lee, and S.H. Jin. **A study on the BCI-Robot assisted stroke rehabilitation framework using brain hemodynamic signals.** In *the 9th Int Conference on Ubiquitous Robots and Ambient Intelligence*, IEEE proceedings, Daejon, Korea, November 26-29 2012.

[32] B. Abibullaev, J. An, S.H.Lee, S.H. Jin, J.I Moon, H.J. Lee. **Decoding of Hemodynamic Brain Responses for Brain-Computer Interfaces via Ensemble Support Vector Learning.** In *the Human-Computer Interaction Conference*, Gangwon-Do, Korea, January 11-13, 2012.

[33] B. Abibullaev, W.S. Kang, S.H. Lee, J. An, and H.D. Seo. **Near-Infrared Spectroscopy in the analysis of functional brain activity during cognitive tasks.** In *2010 IEEE Sensors Conference*, Hawaii, United States, November 1-4, 2010.

[34] B. Abibullaev, W.S. Kang, S.H. Lee, and J. An. **Recognition of brain hemodynamic mental response for brain-computer interfaces.** In *International Conference on Control Automation and Systems*, IEEE proceedings, Gyeonggi-do, Korea, October 27-30 2010.

[35] S.H. Lee, B. Abibullaev, W.S. Kang, and J. An. **analysis of attention deficit hyperactivity disorder in EEG using wavelet transform and self-organizing maps.** In *International Conference on Control Automation and Systems*, IEEE proceedings, Gyeonggi-do, Korea, October 27-30 2010.

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Mentoriship

Current Students:

- Supervisor: Zhenis Otarbay, Ph.D. Student (1st year), Dissertation: "**Design and Evaluation of Brain-Machine Interfaces for Upper-limb Stroke Rehabilitation**"

- Co-Supervisor: Irina Fedorova, Ph.D. Student (3rd year), Supervisor: Prof. Amin Zollanvari, Dissertation: "**Robust Data-driven Predictive Models for Brain-Computer Interfaces**"

- Co-Supervisor: Inara Tusseyava, Ph.D. Student (3rd year), Supervisor: Prof. Matteo Rubagotti, Dissertation: "**Perceived Safety in Human-Robot Interaction**"

- Supervisor: Kuanysh Kudaibergenov, M.Sc. Student (2nd year), Thesis: **"Subject to Subject Transfer Learning for BCIs using Deep Transformers Networks"**

- Supervisor: Kulyan Karipbayeva, M.Sc. Student (2nd year), Thesis: "**Non-Invasive Brain-Computer Interfaces for Speech Synthesis**"

- Supervisor: Aigerim Keutayeva, M.Sc. Student (1st year), Thesis: "**Robust Subject Independent BCIs using Attention Mechanism based CNNs**"

- Co-Supervisor: Vladislav Mun, M.Sc. Student (2nd year), Co-Supervisor: Prof. Siamac Fazli, Thesis: "**Explainable Deep Learning for Brain-Computer Interfaces**"

Past Students:

- Supervisor: Alen Kabdislam, M.Sc. Student (June 2020), Thesis: "**EEG Decoding and Classification using Deep Learning with Convolutional Neural Networks**" (currently Senior Data Scientist at First Heartland Jusan Bank)

- Co-Supervisor: Mukhamet Nurpeiissov, M.Sc. Student (2nd year), Co-Supervisor: Prof. Tohid Alizadeh, Thesis: "**Robot Learning and human-robot interaction framework based on programming by demonstration**" (currently Data Scientist, Institute of Smart Systems and Artificial Intelligence, Nazarbayev University)

- Superved more than 14 undergraduate students formally through graduation projects or informally interested in working with me. Most of these supervised projects led to the 13 different publications either in journals and conferences in the field of Brain-Computer Interfaces.